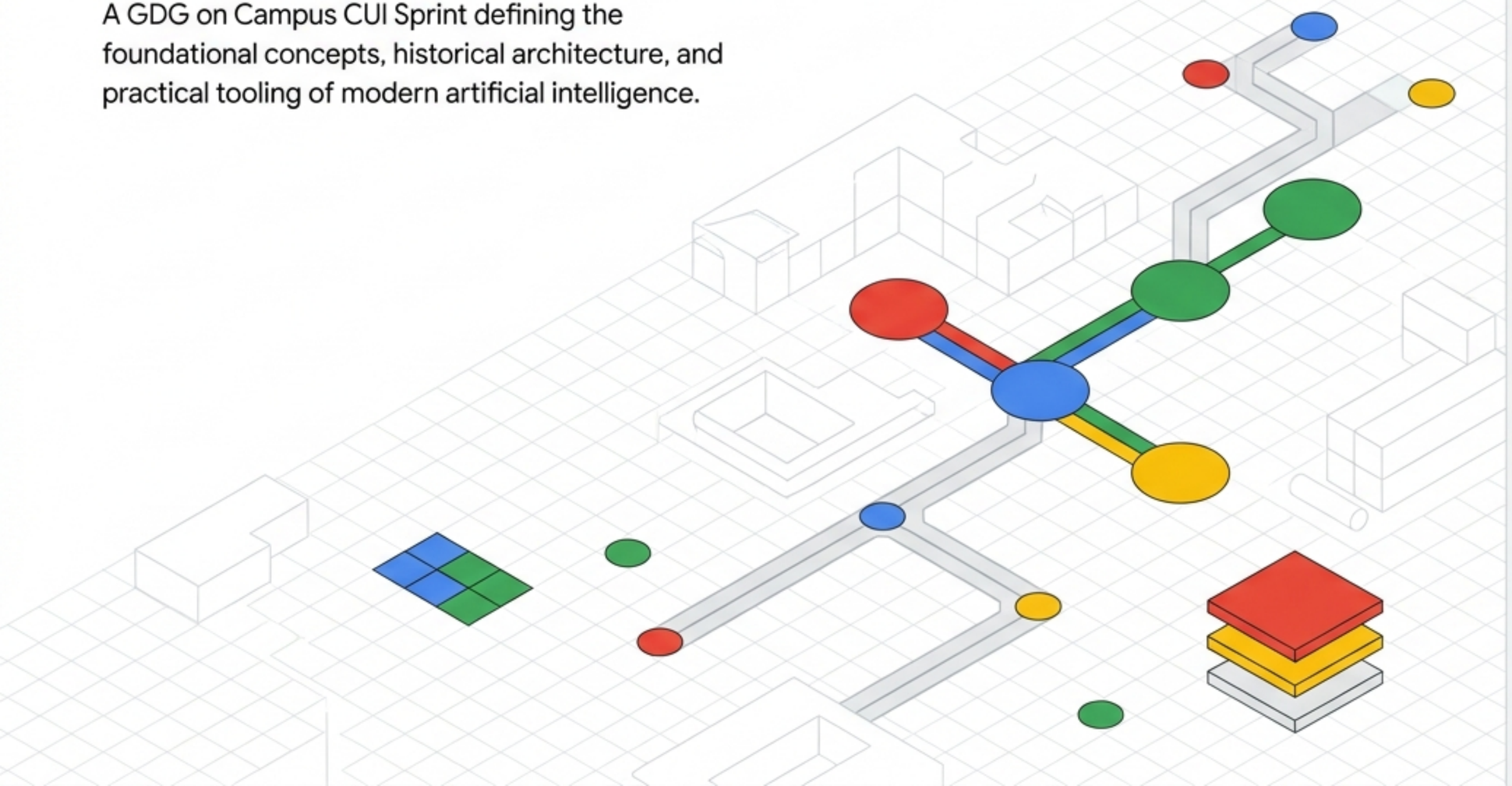
Mapping the Landscape of Generative AI

A GDG on Campus CUI Sprint defining the foundational concepts, historical architecture, and practical tooling of modern artificial intelligence.



INSTRUCTOR NOTES

Session 1 Setup

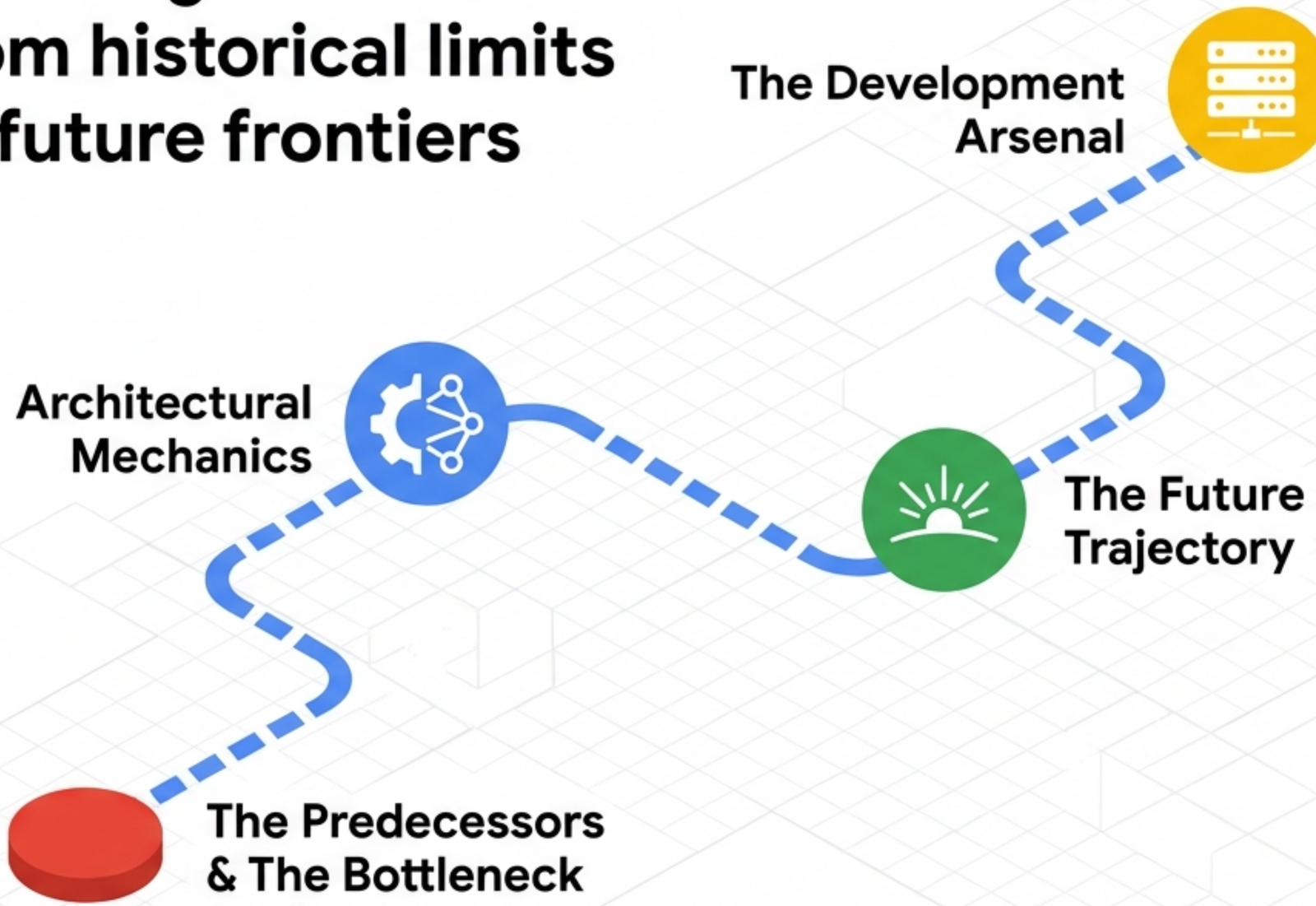
Welcome to the Generative AI sprint.

Today is not about writing dense code. It is about building an intuitive compass.

By the end of this session, the buzzwords surrounding GenAI will be demystified and grounded in practical concepts, preparing you for the hands-on coding sessions to come.

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Charting the course from historical limits to future frontiers



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The Navigator's Guide

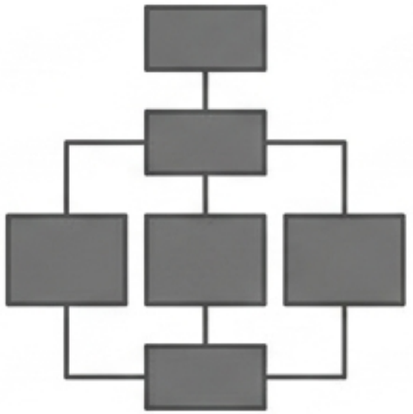
We are building this foundation systematically. We will start by looking at the AI technologies that came before and why they hit a wall.

Then, we trace the historical breakthroughs that made generation possible, peek under the hood of neural networks, forecast where this tech is headed next, and finally map out the exact tools we will use in future sessions.

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The architectural predecessors: Rules and Predictions

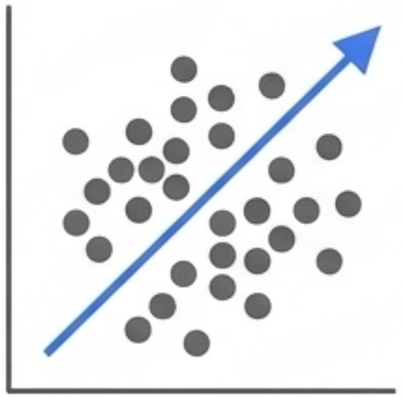
Symbolic AI (1980s-1990s)



Expert Systems

If-Then-Else logic bounded entirely by human-coded rules. High precision, zero adaptability.

Predictive ML (2000s-2010s)



Discriminative ML

Learning patterns from historical data to classify, cluster, or predict future states based on existing distributions.

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Before Generation

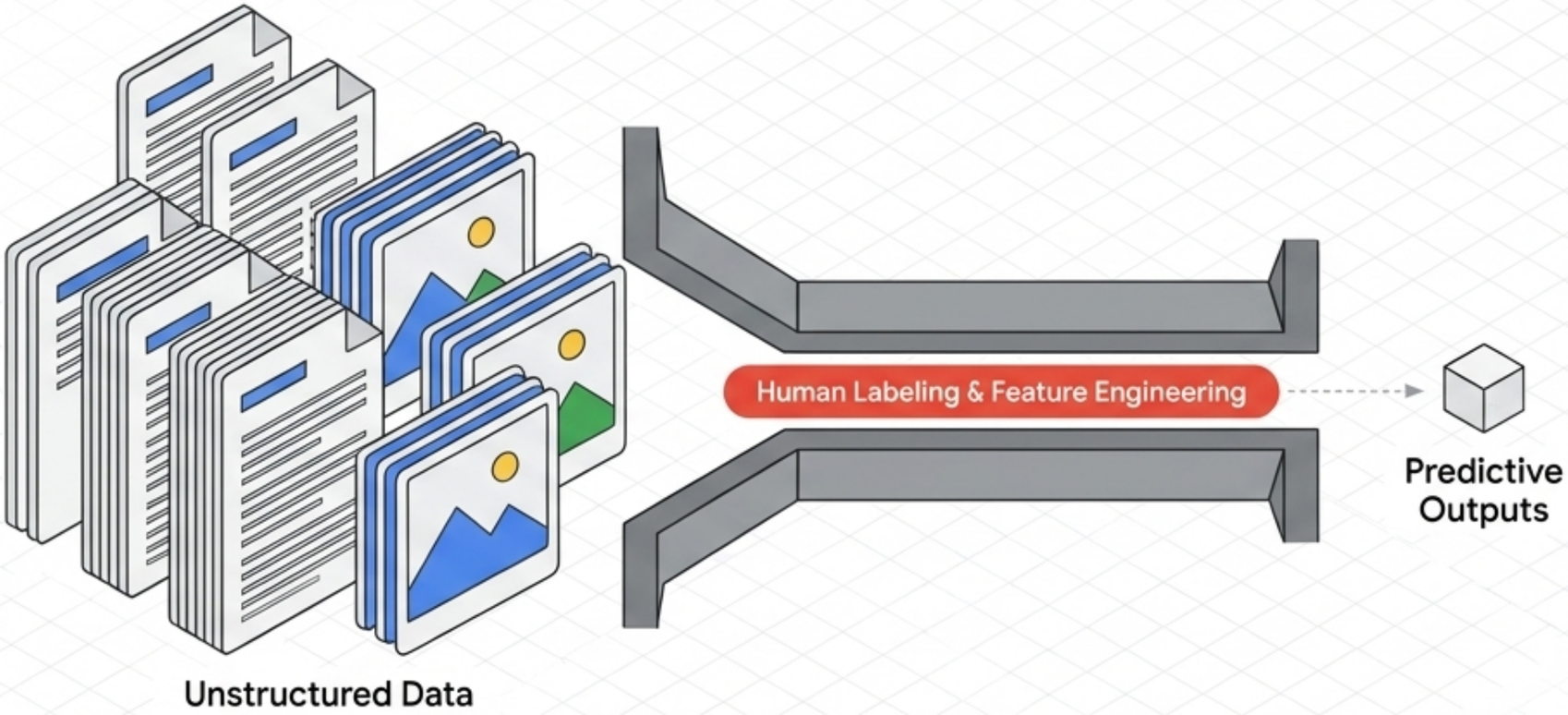
To understand GenAI, we must look at what we built first.

Decades ago, AI was Symbolic—just massive flowcharts of human-written rules. It couldn't handle ambiguity.

Then came Predictive Machine Learning. Instead of writing the rules, we fed the machine data to find the patterns. But even predictive ML had a ceiling: it could only analyze what already existed.

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The creation bottleneck:
Why predictive models hit a ceiling



Predictive ML scales linearly with human effort. To scale exponentially, the system needed the ability to synthesize unstructured data and generate novel outputs without explicit human labeling.

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The Need for Synthesis

Why did we need Generative AI? Because of the bottleneck of creation.
Predictive AI required massive, clean, human-labeled datasets.

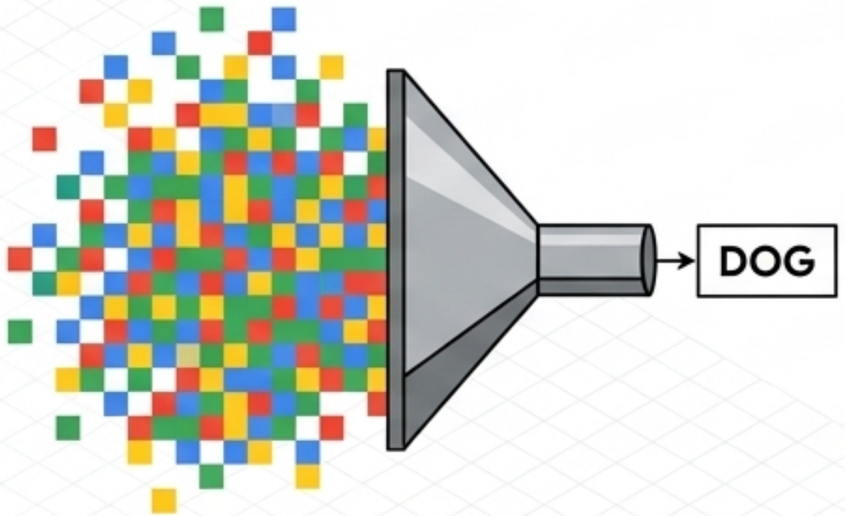
To scale, we needed AI that didn't just label data, but understood the underlying distribution of the data well enough to create it.

We needed a system that could move from mere analysis to active synthesis.

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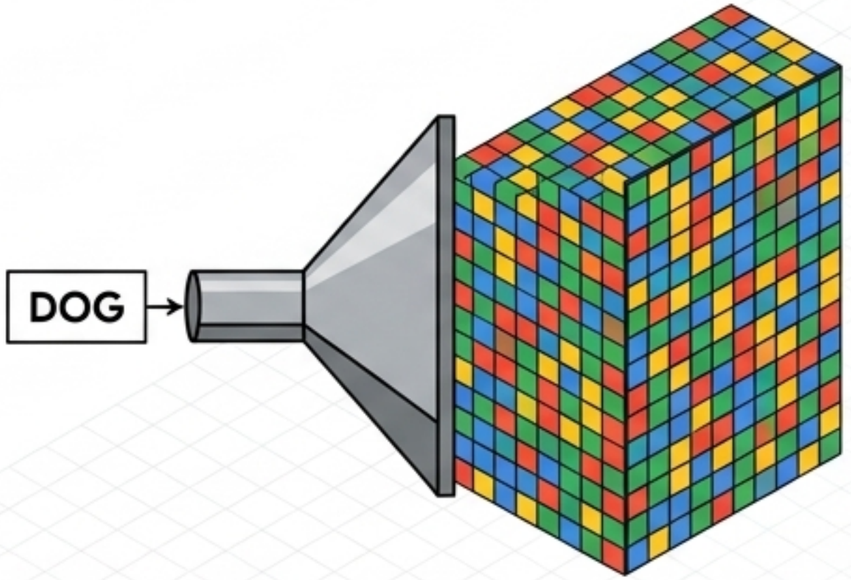
The paradigm shift from analytical critics to generative artists

Discriminative AI (The Critic)



Function: Collapses data to a label.
Primary Value: Analysis & Categorization.

Generative AI (The Artist)



Function: Expands a label into complex data.
Primary Value: Creation & Synthesis.

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Overcoming the Bottleneck

This is how we broke through the bottleneck.

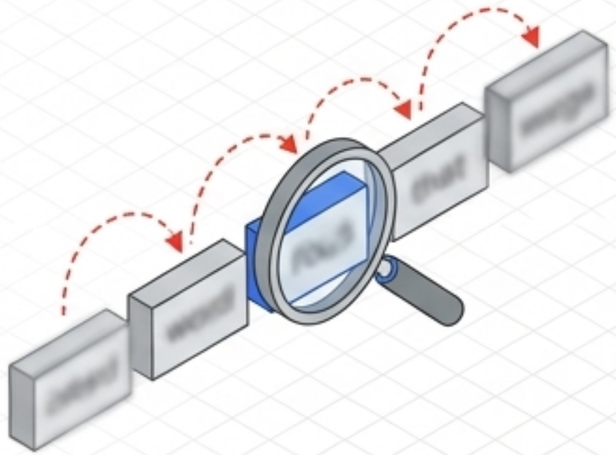
Think of traditional AI as a harsh art critic: it looks at a complex painting and reduces it to a single label ('dog').

Generative AI flips this architecture. It is the artist. You give it the simple label ('dog'), and it expands that label, generating the complex, high-dimensional output (the pixels, the textures, the lighting).

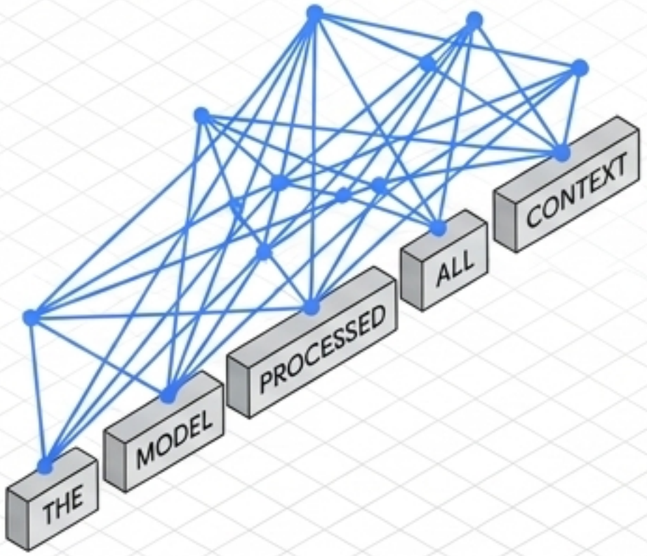
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The architectural evolution that unlocked generative capabilities

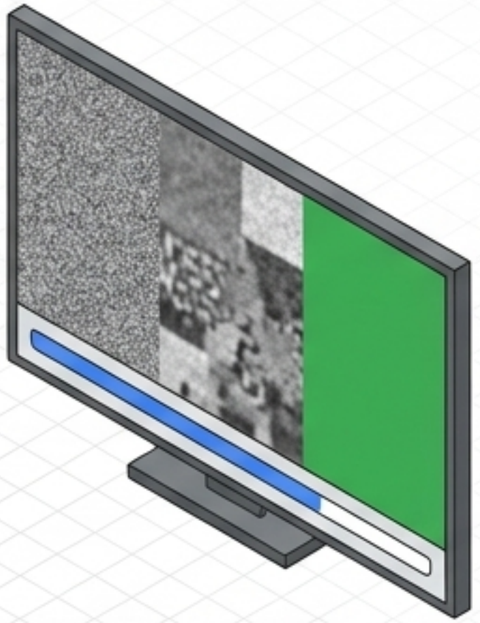
Pre-2017: Sequential Era (RNNs)



2017: Attention Revolution (Transformers)



2020s: Era of Scale (Diffusion & LLMs)



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Historical Milestones

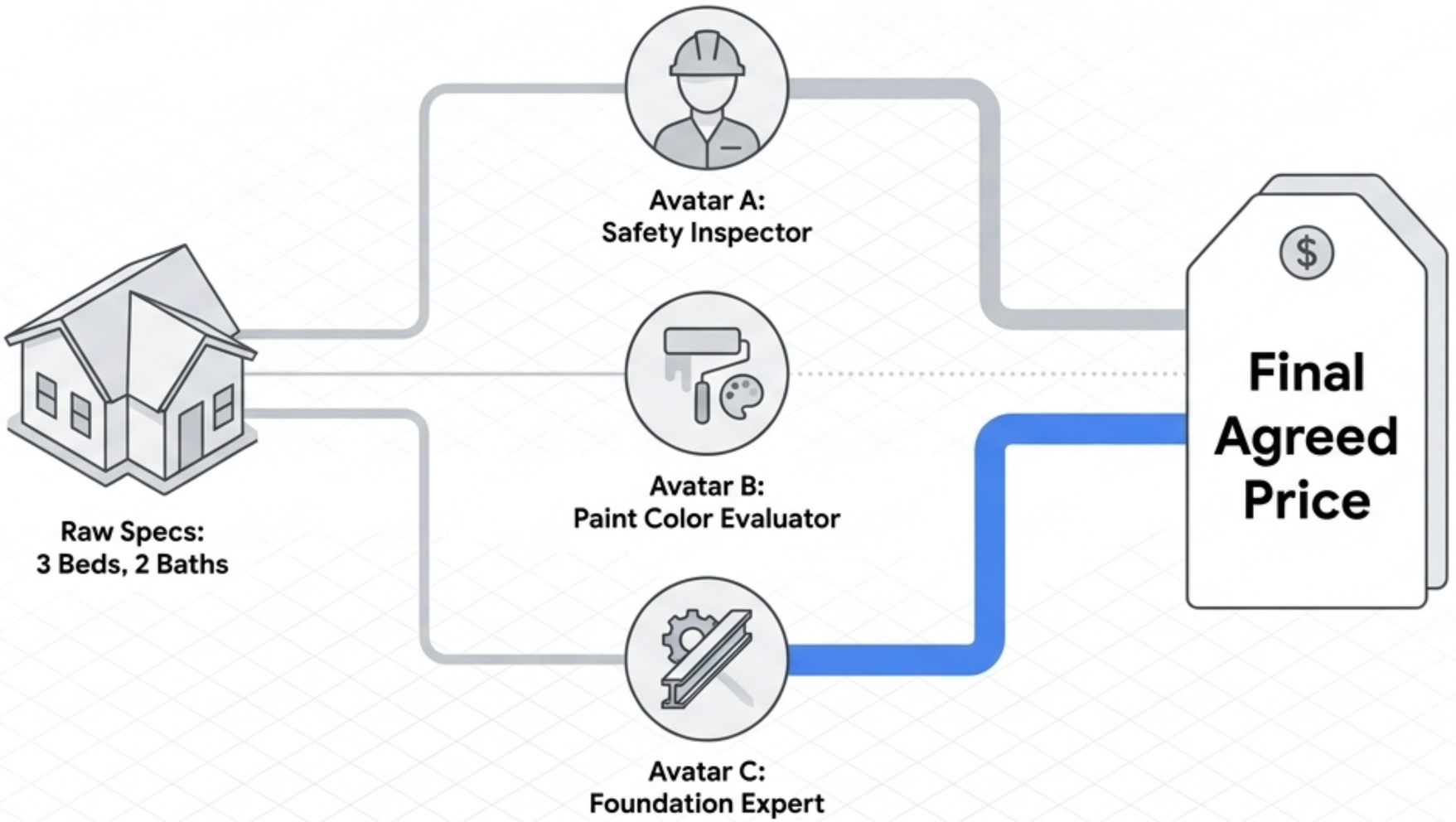
Recurrent Neural Networks (RNNs) read like a human with a poor memory—one word at a time, often forgetting the start of a sentence by the end.

The 2017 Transformer breakthrough introduced Attention, allowing the model to process massive context simultaneously.

Modern Diffusion models work entirely differently, iteratively clearing static from a canvas until a perfect image remains.

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Modeling foundational neural networks as a real estate appraisal committee



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The Mechanics of Trust

To understand these modern architectures, we have to look at their base unit: the neural network.

Think of it as a highly structured committee.

Raw data enters, and hidden layers (sub-committees) evaluate specific features.

The magic lies in the Weights. Weights are just mathematical trust.

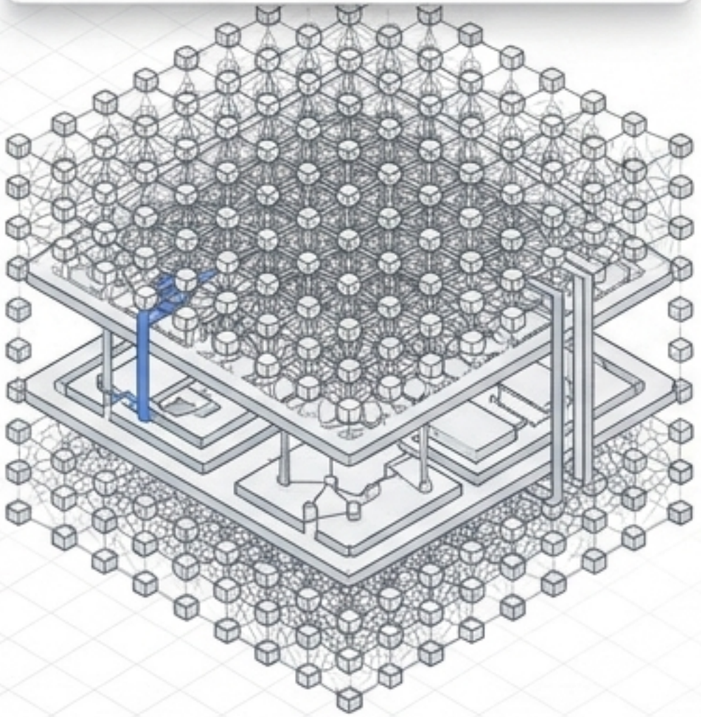
The final output relies heavily on the foundation expert's assessment (thick line) and ignores the paint color evaluator (dotted line).

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Scaling the appraisal committee to comprehend human language

Large Language Models (LLMs) operate on the exact same principles of hidden layers and weights, simply scaled to billions of parameters to evaluate language instead of real estate.





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The Synthesis Moment

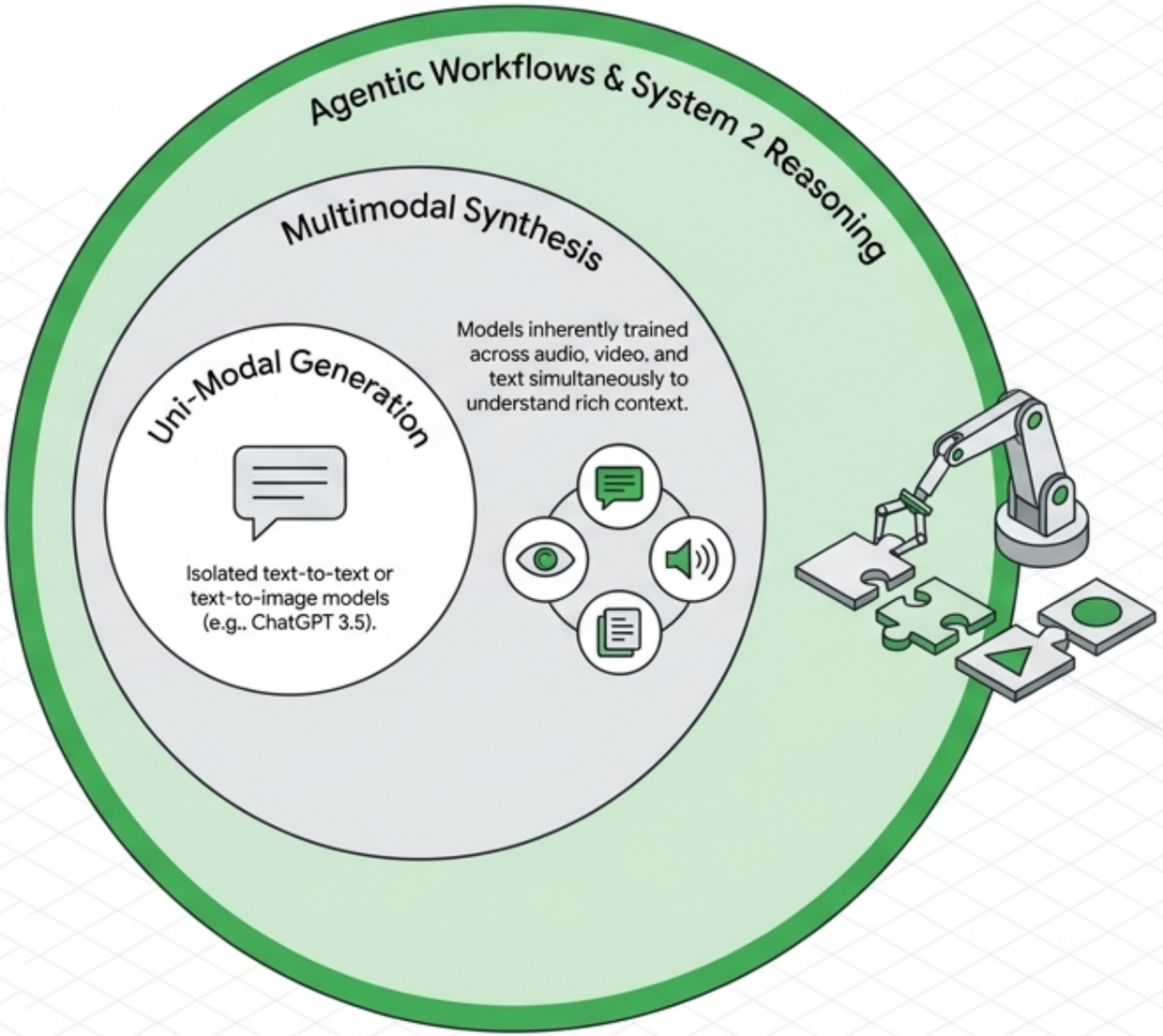
The massive, intimidating LLMs making headlines aren't magic. They are exactly the same as our real estate committee.

Instead of evaluating house specs to predict a price, they evaluate previous words to predict the next word.

The sheer scale of this committee is what allows for the generation of complex, human-like text.

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The expanding horizon: From text generation to autonomous action



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The Trajectory

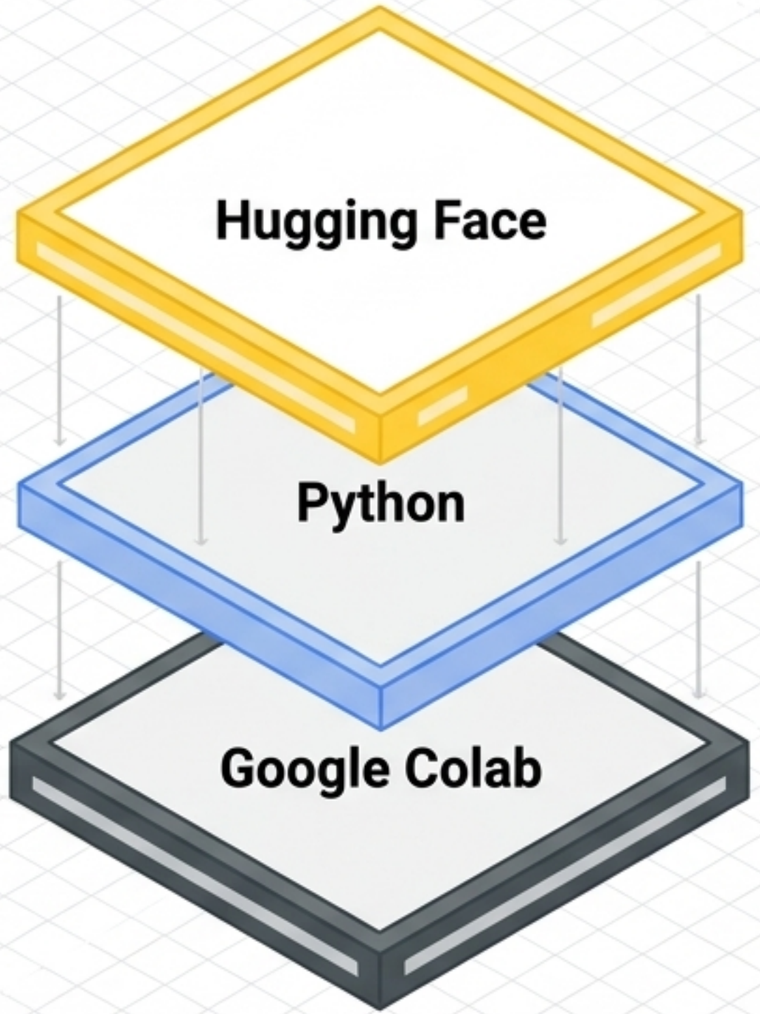
Where is this headed?

We are moving rapidly outward. We started with isolated text generators.
Now, we are in the era of multimodality—models that can see, hear, and speak simultaneously.

The immediate future, the frontier you will be building for, is Agentic AI. These are systems that don't just generate text, but can reason through a problem, create a plan, and execute actions across other software tools autonomously.

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Structuring the modern generative AI development stack



The Repository.
Pre-trained models and datasets ready for deployment.

The Language.
The syntactic glue that binds data to the models.

The Engine.
Cloud-based Jupyter notebooks providing free GPU access.

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The Practical Arsenal

To build for that frontier, you need the right tools. This is your tech stack for the coming weeks.

Google Colab provides the raw cloud computing power so your local machine doesn't have to.

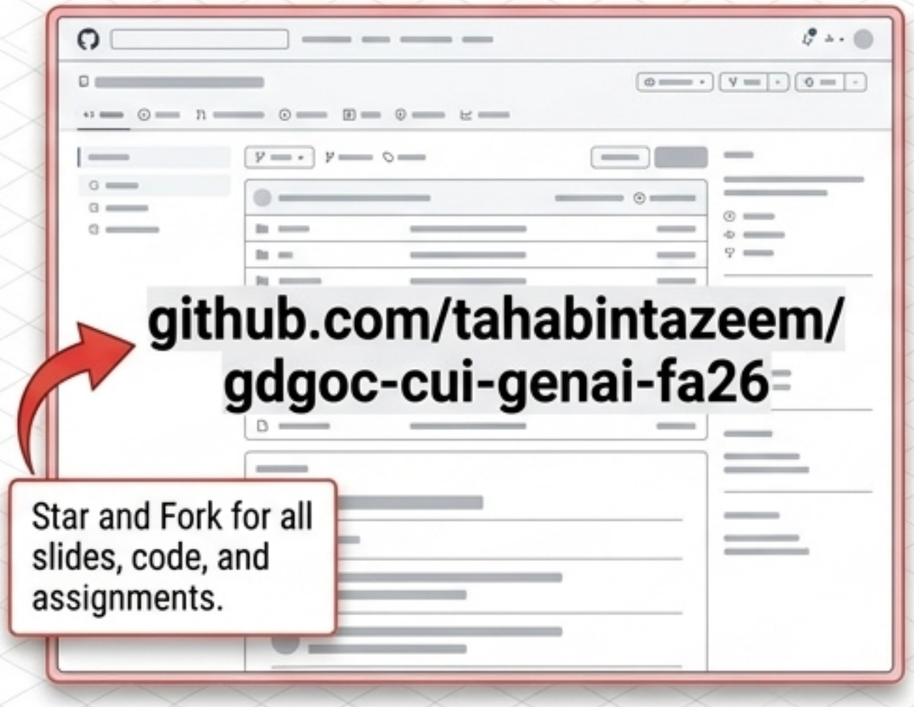
Hugging Face acts as our library of pre-trained models.

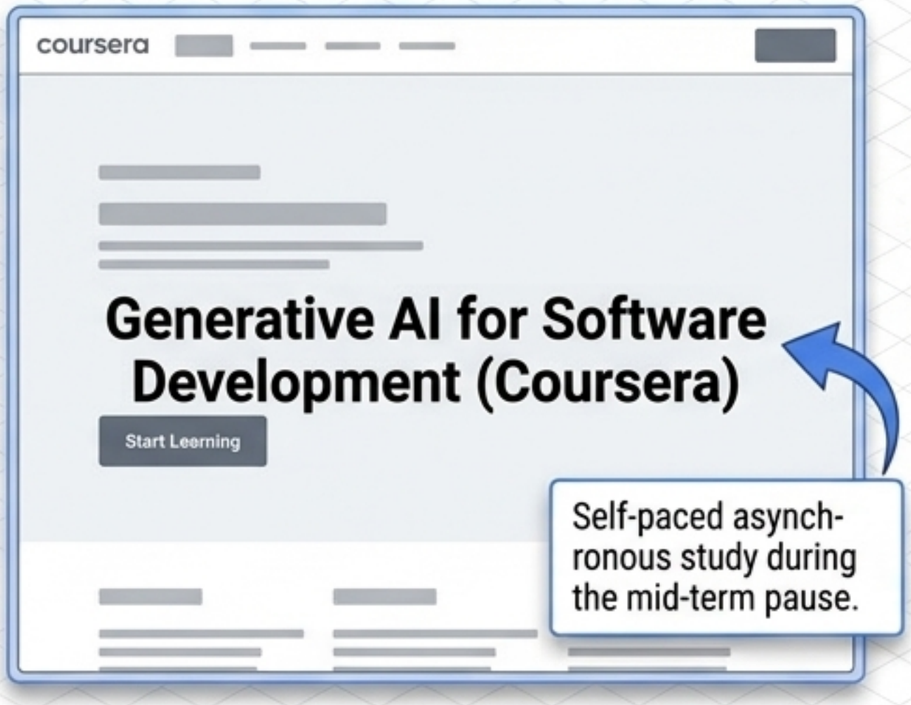
Python is the language we use to command them.

We will learn to orchestrate all three practically.

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Sustaining development momentum through the mid-term break





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Next Steps & Resources

We have a two-week break for mid-terms. Do not lose the foundation we built today.

The GitHub repository is our ultimate source of truth—bookmark it now.

During the break, dive into the Coursera module at your own pace.

Watch the Discord and Whats-App for updates, and we will return ready to write code.

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